

Indoor Substation Maintenance Checklist

PBS Name	:		Capacity (MVA)	:	
Substation Name	:		Date	:	

33/11 KV Power Transformer

General Information:

Manufacturer Name:	Serial No:
Capacity (MVA):	Peak Load (MW):

Visual Inspection:

Oil level	:	☐ OK	☐ Not OK	Grounding	:	☐ OK	☐ Not OK
Oil leakage	:	☐ Yes	☐ No	Condition of bushing	:	☐ OK	☐ Not OK
Silica gel changed	:	☐ Yes	☐ No	Connection tightness	:	☐ OK	☐ Not OK
Riser connections opened and rejoined	:	☐ Yes	☐ No	Dust cleaned (MK and XR upper surface)	:	☐ Yes	☐ No

Functionality Checking:

Buchholz alarm	:	☐ Yes	☐ No	WTI alarm	:	☐ Yes	☐ No
Buchholz trip	:	☐ Yes	☐ No	WTI trip	:	☐ Yes	☐ No
OTI alarm	:	☐ Yes	☐ No	Cooling fan manual operation	:	☐ Yes	☐ No
OTI trip	:	☐ Yes	☐ No	Cooling fan remote operation	:	☐ Yes	☐ No
PRD trip	:	☐ Yes	☐ No	Cooling fan auto operation	:	☐ Yes	☐ No

Testing:

Insulation resistance	:	☐ Satisfactory	☐ Not Satisfactory
Di-electric strength of oil	:	☐ Satisfactory	☐ Not Satisfactory
Transformer turns ratio at each tap position	:	☐ Satisfactory	☐ Not Satisfactory

Comment: Test reports attached

OLTC & RTCC

OLTC Manufacturer	:		RTCC Manufacturer	:	
OLTC Model & Type	:		RTCC Serial No	:	
OLTC Serial No.	:			:	

Visual Inspection:

OLTC oil level	:	☐ OK	☐ Not OK	Dust cleaned (RTCC panel and Motor Drive Unit)	:	☐ OK	☐ Not OK
Tap position indicator	:	☐ OK	☐ Not OK		:		
Tightness of connections	:	☐ OK	☐ Not OK		:		

Functionality Checking:

OLTC local operation	:	☐ Yes	☐ No	OLTC Buchholz alarm	:	☐ Yes	☐ No
OLTC remote operation	:	☐ Yes	☐ No	OLTC Buchholz trip	:	☐ Yes	☐ No
OLTC auto operation	:	☐ Yes	☐ No	OLTC PRD trip	:	☐ Yes	☐ No

Comment:

33 kV VCB & CRP Panel

VCB Manufacturer	:		CRP Manufacturer	:	
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Visual Inspection:

Position	:	33 KV Line Incomer	33 KV Transformer Incomer
Model & Manufacturing year	:		
VCB Serial No.	:		
CRP Serial No.	:		
VCB pole condition	:	☐ OK ☐ Not OK	☐ OK ☐ Not OK
Riser connection opened & rejoined	:	☐ Yes ☐ No	☐ Yes ☐ No
Mechanical joints lubricated	:	☐ Yes ☐ No	☐ Yes ☐ No
Device in CRP (TCS, master trip, meters)	:	☐ OK ☐ Not OK	☐ OK ☐ Not OK
PT voltage in energy meter	:	☐ OK ☐ Not OK	☐ OK ☐ Not OK
Connections tightness	:	☐ OK ☐ Not OK	☐ OK ☐ Not OK
Dust cleared (VCB, control cubicle, CRP, MK)	:	☐ Yes ☐ No	☐ Yes ☐ No

Functionality Checking:

VCB local closing	:	☐ Yes ☐ No	☐ Yes ☐ No
VCB remote closing	:	☐ Yes ☐ No	☐ Yes ☐ No
VCB local tripping	:	☐ Yes ☐ No	☐ Yes ☐ No
VCB remote tripping	:	☐ Yes ☐ No	☐ Yes ☐ No
Protection tripping	:	☐ Yes ☐ No	☐ Yes ☐ No
Master trip operation	:	☐ Yes ☐ No	☐ Yes ☐ No
TCS-1 relay healthy	:	☐ Yes ☐ No	☐ Yes ☐ No
TCS-2 relay healthy	:	☐ Yes ☐ No	☐ Yes ☐ No
Automatic spring charge	:	☐ Yes ☐ No	☐ Yes ☐ No
VCB indications	:	☐ Yes ☐ No	☐ Yes ☐ No
CRP indications	:	☐ Yes ☐ No	☐ Yes ☐ No
Heater operation	:	☐ Yes ☐ No	☐ Yes ☐ No

Testing:

Insulation resistance of VCB	:	☐ Satisfactory ☐ Not Satisfactory	☐ Satisfactory ☐ Not Satisfactory
Contact resistance of VCB	:	☐ Satisfactory ☐ Not Satisfactory	☐ Satisfactory ☐ Not Satisfactory
Opening time of VCB	:	☐ Satisfactory ☐ Not Satisfactory	☐ Satisfactory ☐ Not Satisfactory

Comment: Test reports attached

Overcurrent & Earth Fault Relay

Position	:	33 KV Line Incomer	33 KV Transformer Incomer
Manufacturer & Model	:		
Country of Origin	:		
Serial No.	:		
Watchdog	:	☐ OK ☐ Not OK	☐ OK ☐ Not OK
Auxiliary voltage	:	☐ OK ☐ Not OK	☐ OK ☐ Not OK
Date and time	:	☐ OK ☐ Not OK	☐ OK ☐ Not OK
Relay display	:	☐ OK ☐ Not OK	☐ OK ☐ Not OK
Relay buttons working	:	☐ Yes ☐ No	☐ Yes ☐ No
Relay terminals cleaned	:	☐ Yes ☐ No	☐ Yes ☐ No
All connections tightness	:	☐ OK ☐ Not OK	☐ OK ☐ Not OK
Protection Setting Checking	:	☐ Yes ☐ No	☐ Yes ☐ No

33 kV Line Incomer Current Transformer

Manufacturer	:		Connected CT Ratio	:	
Rated CT Ratio	:		Rated Burden	:	

Visual Inspection:

Serial No.	:			
Condition of bushing	:	☐ OK ☐ Not OK	☐ OK ☐ Not OK	☐ OK ☐ Not OK
Oil leakage	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Primary connections opened and rejointed	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Secondary connections tightened	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Star-point grounded	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Dust cleared(MK, Secondary terminal box)	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Unused core shorted	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No

Testing:

Insulation resistance	:	☐ Satisfactory ☐ Not Satisfactory	☐ Satisfactory ☐ Not Satisfactory	☐ Satisfactory ☐ Not Satisfactory
Excitation test	:	☐ Satisfactory ☐ Not Satisfactory	☐ Satisfactory ☐ Not Satisfactory	☐ Satisfactory ☐ Not Satisfactory

Comment: Test reports attached

33 kV Line Potential Transformer

Manufacturer	:		PT Ratio	:	
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Visual Inspection:

Serial No.	:			
Condition of bushing	:	☐ OK ☐ Not OK	☐ OK ☐ Not OK	☐ OK ☐ Not OK
Oil leakage	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Primary connections opened and rejointed	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Secondary connections tightened	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Neutral grounded	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Dust cleared(Junction box, Secondary terminal box)	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No

Testing:

Insulation resistance	:	☐ Satisfactory ☐ Not Satisfactory	☐ Satisfactory ☐ Not Satisfactory	☐ Satisfactory ☐ Not Satisfactory
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Comment: Test reports attached

33 kV Transformer Incomer Current Transformer

Manufacturer	:		Connected CT Ratio	:	
Rated CT Ratio	:		Rated Burden	:	

Visual Inspection:

Serial No.	:			
Condition of bushing	:	☐ OK ☐ Not OK	☐ OK ☐ Not OK	☐ OK ☐ Not OK
Oil leakage	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Primary connections opened and rejointed	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Secondary connections tightened	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Star-point grounded	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Dust cleared (MK, Secondary terminal box)	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Unused core shorted	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No

Testing:

Insulation resistance	:	☐ Satisfactory ☐ Not Satisfactory	☐ Satisfactory ☐ Not Satisfactory	☐ Satisfactory ☐ Not Satisfactory
Excitation test	:	☐ Satisfactory ☐ Not Satisfactory	☐ Satisfactory ☐ Not Satisfactory	☐ Satisfactory ☐ Not Satisfactory

Comment: Test reports attached

33 kV Bus Potential Transformer

Manufacturer	:		PT Ratio	:	
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Visual Inspection:

Serial No.	:			
Condition of bushing	:	☐ OK ☐ Not OK	☐ OK ☐ Not OK	☐ OK ☐ Not OK
Oil leakage	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Primary connections opened and rejointed	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Secondary connections tightened	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Neutral grounded	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Dust cleared(Junction box, Secondary terminal box)	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No

Testing:

Insulation resistance	:	☐ Satisfactory ☐ Not Satisfactory	☐ Satisfactory ☐ Not Satisfactory	☐ Satisfactory ☐ Not Satisfactory
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Comment: Test reports attached

Differential Relay

Manufacturer	:		Country of Origin	:	
Model	:		Serial No	:	

Visual Inspection:

Watchdog	:	☐ OK ☐ Not OK	Relay buttons working	:	☐ Yes ☐ No
Auxiliary voltage	:	☐ OK ☐ Not OK	Relay terminals cleaned	:	☐ Yes ☐ No
Date and time	:	☐ OK ☐ Not OK	All connections tightness	:	☐ OK ☐ Not OK
Relay display	:	☐ OK ☐ Not OK			

Comment:

33/0.415 kV Auxiliary Transformer

Manufacturer	:		Serial No	:	
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Visual Inspection:

HV and LV bushing	:	☐ OK	☐ Not OK	Silica gel	:	☐ OK	☐ Not OK
HV connections	:	☐ OK	☐ Not OK	LV connections	:	☐ OK	☐ Not OK

Testing:

Insulation Resistance	:	☐ Satisfactory	☐ Not Satisfactory
Di-electric strength of oil	:	☐ Satisfactory	☐ Not Satisfactory
Transformer turns ratio	:	☐ Satisfactory	☐ Not Satisfactory

Comment: Test reports attached**AC & DC Distribution Panel**

Manufacturer	:	
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Visual Inspection:

All meters measurement	:	☐ OK	☐ Not OK	Proper marking of MCBs	:	☐ OK	☐ Not OK
Panels cleanliness	:	☐ OK	☐ Not OK	Tightness of connections	:	☐ OK	☐ Not OK

Functionality Checking:

Indication lamps	:	☐ Yes	☐ No	MCB and MCCBs operation	:	☐ Yes	☐ No
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Battery & Battery Charger

Battery Manufacturer	:		Charger Manufacturer	:	
Charger Model	:		Charger Serial No	:	

Visual Inspection:

Electrolyte level	:	☐ OK	☐ Not OK	Battery cells cleaned	:	☐ OK	☐ Not OK
Battery terminal connections tightened	:	☐ Yes	☐ No	Battery terminal points cleaned	:	☐ Yes	☐ No
Battery overheated	:	☐ Yes	☐ No	Charger input	:	☐ OK	☐ Not OK
Charger output	:	☐ OK	☐ Not OK	Charger output when battery isolated	:	☐ OK	☐ Not OK

Functionality Checking:

Float charging	:	☐ Yes	☐ No	Boost charging	:	☐ Yes	☐ No
Manual mode	:	☐ Yes	☐ No	Auto mode	:	☐ Yes	☐ No
Indication bulbs	:	☐ Yes	☐ No	Alarms	:	☐ Yes	☐ No

Testing:

Specific gravity of electrolyte	:	☐ Satisfactory	☐ Not Satisfactory
Battery fully discharged	:	☐ Satisfactory	☐ Not Satisfactory

Comment: Test reports attached**Grounding & Miscellaneous****Visual Inspection:**

All equipment properly grounded	:	☐ OK	☐ Not OK
Positive/negative earth fault alarm in battery charger panel	:	☐ Yes	☐ No
Air Condition in control room	:	☐ OK	☐ Not OK

Testing:

Earth resistance at various location of the substation	:	☐ Satisfactory	☐ Not Satisfactory
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11 kV AIS Switchgear Panel

Position	:		Model	:	
Manufacturer	:		Serial No	:	

11 kV VCB

Position	:		Manufacturer	:	
Serial No	:			:	

Visual Inspection:

VCB rack in & rack out	:	☐ OK ☐ Not OK	Dust cleared (inside VCB panel, mechanical chamber)	:	☐ Yes ☐ No
Mechanical joints lubricated	:	☐ Yes ☐ No			
Connections tightness	:	☐ OK ☐ Not OK			

Functionality Checking:

VCB local closing	:	☐ Yes ☐ No	Automatic spring charge	:	☐ Yes ☐ No
VCB local tripping	:	☐ Yes ☐ No	Indications	:	☐ Yes ☐ No
Protection tripping	:	☐ Yes ☐ No			

Testing:

Insulation Resistance	:	☐ Satisfactory ☐ Not Satisfactory
Contact resistance	:	☐ Satisfactory ☐ Not Satisfactory
Opening time	:	☐ Satisfactory ☐ Not Satisfactory

Comment: Test reports attached

CRP

Manufacturer	:		Serial No	:	
VCB Serial No	:		CRP	:	

Visual Inspection:

Device in CRP (TCS, master trip, meters)	:	☐ OK ☐ Not OK	Dust cleared (Terminal blocks, contact points)	:	☐ Yes ☐ No
PT voltage in energy meter	:	☐ OK ☐ Not OK	Connections tightness	:	☐ OK ☐ Not OK

Functionality Checking:

VCB remote closing	:	☐ Yes ☐ No	TCS-1 relay healthy	:	☐ Yes ☐ No
VCB remote tripping	:	☐ Yes ☐ No	TCS-2 relay healthy	:	☐ Yes ☐ No
Protection tripping	:	☐ Yes ☐ No	Indications	:	☐ Yes ☐ No
Master trip operation	:	☐ Yes ☐ No	Heater operation	:	☐ Yes ☐ No

Overcurrent & Earth Fault Relay

Position	:		Country of Origin	:	
Manufacturer & Model	:		Serial No	:	

Visual Inspection:

Watchdog	:	☐ OK ☐ Not OK	Relay buttons working	:	☐ Yes ☐ No
Auxiliary voltage	:	☐ OK ☐ Not OK	Relay terminals cleaned	:	☐ Yes ☐ No
Date and time	:	☐ OK ☐ Not OK	All connections tightness	:	☐ OK ☐ Not OK
Relay display	:	☐ OK ☐ Not OK			

Protection Setting Checking:

Protection setting as per BREB SPD's guideline	:	☐ OK ☐ Not OK
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11 kV Current Transformer

Manufacturer	:		Connected CT Ratio	:	
Rated CT Ratio	:				

Visual Inspection:

Serial No.	:			
Primary connections opened and rejoined	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Secondary connections tightened	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Star-point grounded	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Dust cleared(MK, Secondary terminal box)	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Unused core shorted	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No

Testing:

Insulation resistance	:	☐ Satisfactory ☐ Not Satisfactory	☐ Satisfactory ☐ Not Satisfactory	☐ Satisfactory ☐ Not Satisfactory
Excitation test	:	☐ Satisfactory ☐ Not Satisfactory	☐ Satisfactory ☐ Not Satisfactory	☐ Satisfactory ☐ Not Satisfactory

Comment: Test reports attached

Bus-bar

Visual Inspection:

Bus-bar condition	:	☐ OK ☐ Not OK	Connection tightness	:	☐ OK ☐ Not OK
Cable connection opened & rejoined	:	☐ OK ☐ Not OK	Dust cleared	:	☐ Yes ☐ No

Testing:

Insulation resistance	:	☐ Satisfactory ☐ Not Satisfactory
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Comment: Test reports attached

11 kV Potential Transformer

Manufacturer	:		PT Ratio	:	
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Visual Inspection:

Serial No.	:			
Primary connections opened and rejoined	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Secondary connections tightened	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Neutral grounded	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Dust cleared(Junction box, Secondary terminal box)	:	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No

Testing:

Insulation resistance	:	☐ Satisfactory ☐ Not Satisfactory	☐ Satisfactory ☐ Not Satisfactory	☐ Satisfactory ☐ Not Satisfactory
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Comment: Test reports attached